



DataSynth

Enterprise Synthetic Data Platform

High-Performance Generation for Accounting, Audit & ML

Rust core • Python 1.10 • REST/gRPC API • Audit FSM Engine

Executive Overview

Version 2.2.0

Deterministic, privacy-preserving synthetic data generation
for enterprise finance, compliance testing, and machine learning.

Built with Rust & Python • 20 modular crates • 200 K+ entries/sec

Open-source (Apache-2.0) • Commercial license available

GitHub: <https://github.com/mivertowski/SyntheticData>

Contents

1	Executive Summary	2
1.1	At a Glance	3
2	Platform Capabilities	4
2.1	Enterprise Accounting Simulation	4
2.2	Financial Coherence Engine	4
2.2.1	Manufacturing Cost Flow	4
2.2.2	Treasury & Debt Accounting	4
2.2.3	Tax Integration	5
2.2.4	Payroll ← HR Events	5
2.2.5	Full Financial Statement Package	5
2.2.6	Capstone Coherence Validators	5
2.3	Enterprise Process Chains	6
2.4	Country Pack Architecture	8
2.5	Banking, KYC & AML	8
2.6	Audit Simulation	8
2.7	YAML-Driven Audit FSM Engine	9
2.7.1	Two-Layer Architecture	9
2.7.2	Key Capabilities	9
2.7.3	Audit Path Optimizer	10
2.8	COSO 2013 Internal Control Framework	10
2.9	Accounting & Audit Standards	10
2.10	Process Mining (OCEL 2.0 & XES 2.0)	11
3	AI-Augmented Generation	12
3.1	LLM-Augmented Metadata Enrichment	12
3.2	Diffusion Model Integration	12
3.3	Causal & Counterfactual Generation	12
3.4	Unified Generation Pipeline	13
4	Machine Learning & Analytics	14
4.1	ACFE-Aligned Fraud Taxonomy	14
4.2	Advanced Fraud Modeling	14
4.3	Data Quality Variations	14
4.4	Enterprise Interconnectivity	15
4.5	Pattern & Process Drift	15
4.6	Graph & Network Export	16
5	Privacy & Fingerprinting	17
5.1	Privacy-Preserving Fingerprint Extraction	17
5.2	Privacy Levels	17
5.3	Federated Fingerprinting & Certificates	17
5.4	Formal DP Composition	17
5.5	Fidelity Evaluation	18
6	Architecture	18
6.1	Layered Crate Architecture	18
6.2	Production-Grade Infrastructure	19

7	Industry Presets	20
8	Statistical Distribution Framework	20
9	Use Cases	22
10	Evaluation & Auto-Tuning	23
10.1	ML Benchmarks	23
10.2	Quality Gate Engine	23
11	Compliance & Governance	23
11.1	EU AI Act	23
11.2	Regulatory Compliance Documentation	24
12	DevOps & Deployment	24
12.1	CI/CD Pipeline	24
12.2	Deployment Options	24
12.3	Ecosystem Integrations	24
13	Getting Started	26
13.1	Quick Start (CLI)	26
13.2	Quick Start (Python)	26
13.3	Server Mode	26

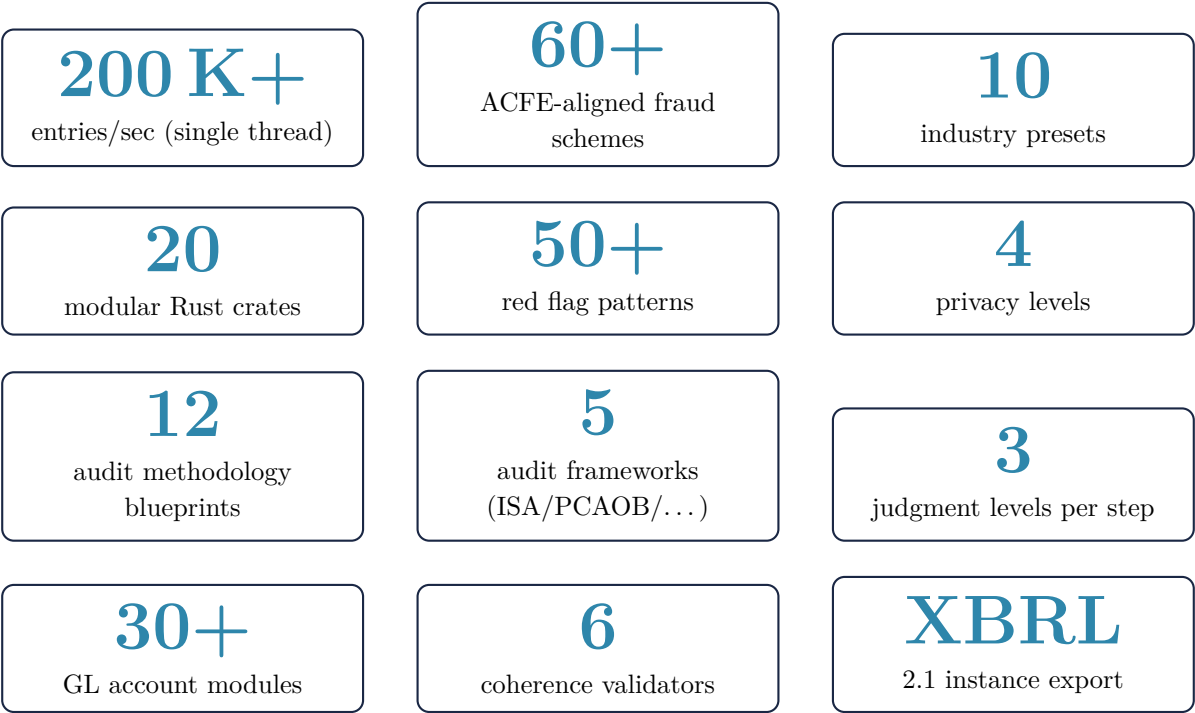
1 Executive Summary

DataSynth is a high-performance synthetic data platform purpose-built for enterprise accounting, audit analytics, and machine learning. Written in Rust for maximum throughput and memory safety — with a full Python wrapper for data-science workflows — it generates realistic, internally coherent financial data that satisfies the statistical and structural properties of real-world enterprise resource planning (ERP) systems.

Why DataSynth?

- **No real data required** — eliminates privacy, regulatory, and procurement barriers to analytics development.
- **Full auditability** — deterministic, seeded generation means every dataset is perfectly reproducible.
- **ML-ready from day one** — ground-truth labels for fraud, anomalies, and data quality ship alongside every record.
- **Domain depth** — covers the full accounting lifecycle: journal entries, document flows, subledgers, FX, intercompany, period close, and banking/AML.
- **Empirically grounded** — statistical distributions for journal entry line items, amount patterns, and temporal volumes are calibrated against empirical research conducted on real-world enterprise datasets, ensuring synthetic output mirrors the structural properties of production ERP data.
- **Methodology-driven audit simulation** — 12 audit methodology blueprints (ISA, IIA-GIAS, PCAOB, AICPA SOC 2, Big 4 firm-style) encoded as YAML-defined finite state machines, producing event-sourced audit trails with judgment-level classification and cross-firm benchmark comparison.

1.1 At a Glance



2 Platform Capabilities

2.1 Enterprise Accounting Simulation

DataSynth generates a complete, internally consistent accounting universe:

Domain	Capabilities
Journal Entries	Balanced debits/credits, Benford-compliant amounts, configurable line-item distributions, SAP ACDOCA format export.
Master Data	Vendors, customers, materials, fixed assets, employees with hierarchies, payment terms, credit ratings, and intercompany flags.
Document Flows	Full Procure-to-Pay (PO → GR → Invoice → Payment) and Order-to-Cash (SO → Delivery → Invoice → Receipt) with three-way match validation, contract price enforcement, and credit memo generation (configurable <code>returns_rate</code> with ARCreditMemo).
Intercompany	Matched IC journal entry pairs, transfer pricing (Cost-Plus, Resale-Minus, CUP), and consolidation elimination entries.
Subledgers	AR/AP open items and aging, fixed asset register with IRS MACRS depreciation tables (3–20 yr recovery periods), inventory positions and movements, GL-to-subledger reconciliation.
FX & Translation	Ornstein–Uhlenbeck exchange rate process, multi-currency trial balance translation (current-rate, temporal, monetary/non-monetary methods with account-type classification), currency translation adjustment entries.
Period Close	Month-end accruals, depreciation runs, year-end closing entries, fiscal period status tracking.

2.2 Financial Coherence Engine

New in v2.2.0 — a cross-module coherence layer that connects manufacturing, treasury, tax, HR, and reporting generators into a unified financial data fabric. Every journal entry traces back to an originating business event, and every financial statement line item reconciles to the underlying GL.

2.2.1 Manufacturing Cost Flow

- **WIP → FG → COGS cycle** — multi-stage cost accumulation with standard cost variance journal entries for material price, labor rate, and overhead volume variances.
- **Warranty provisions** — IAS 37/ASC 450 provisions derived from quality inspection failure rates, with automated accrual and release entries.
- **COGS on delivery** — cost of goods sold recognized at shipment, linked to production unit costs for full cost traceability.

2.2.2 Treasury & Debt Accounting

- **Effective interest method** — debt interest accrual journal entries computed from amortized cost and contractual terms.
- **Hedge mark-to-market** — ASC 815/IFRS 9 routing with cash flow hedge gains/losses posted to OCI and fair value hedge adjustments posted to P&L.

- **Covenant compliance** — real-time evaluation against trial balance ratios (debt/equity, interest coverage, current ratio).
- **Cash pool sweeps** — physical sweep intercompany entries with proper elimination on consolidation.

2.2.3 Tax Integration

- **Income tax provisions** — derived from actual pre-tax income by netting revenue and expense GL accounts, ensuring provision-to-GL reconciliation.
- **Transaction-level VAT/GST** — input and output tax journal entries generated from source documents (vendor invoices, customer invoices, payments).
- **Document-date-aware posting** — tax entries posted on the source document date, avoiding artificial period-end clustering.

2.2.4 Payroll ← HR Events

- **Change-history-driven payroll** — salary adjustments consume `EmployeeChangeHistory` records with mid-period proration for hires, promotions, and terminations.
- **Variance traceability** — payroll period-over-period variances are traceable to specific HR events, supporting audit walkthroughs of compensation changes.

2.2.5 Full Financial Statement Package

- **Enhanced cash flow statement** — operating, investing, and financing items sourced from all active modules (treasury, tax, payroll, manufacturing, project accounting).
- **IFRS vs US GAAP classification** — interest paid/received classified per framework requirements (operating vs. financing).
- **Segment reporting** — operating segments derived from actual entity-level GL data with segment-to-consolidated reconciliation.
- **Data-backed notes** — financial statement notes for inventory, debt maturities, hedge positions, and provisions with traceable source references to underlying records.
- **Dividend lifecycle** — declaration, accrual, and payment journal entries with proper equity and cash flow classification.
- **XBRL 2.1 export** — instance document generation with US GAAP and IFRS taxonomy mapping for regulatory filing.

2.2.6 Capstone Coherence Validators

- **Cash flow reconciliation** — validates that opening cash plus net operating/investing/financing flows equals the closing GL cash balance.
- **Equity rollforward** — confirms opening equity plus net income plus OCI minus dividends equals closing equity.
- **Segment reconciliation** — verifies that segment totals reconcile to consolidated financial statement figures.
- **Trial balance master proof** — asserts that the sum of all posted journal entries equals the closing trial balance for every account.

2.3 Enterprise Process Chains

Updated in v0.7.0 — eleven end-to-end process chains with full referential integrity, temporal coherence, anomaly injection, and OCEL 2.0 process mining integration:

Chain	Capabilities
Source-to-Pay	Spend analysis, sourcing projects, supplier qualification, RFx/bid evaluation, procurement contracts, catalog management, supplier scorecards — with contract-governed PO generation and maverick spend detection.
Hire-to-Retire	Payroll runs with progressive tax bracket computation (country-pack-aware), time entries with overtime modeling, expense reports with policy violation detection, and payroll journal entry generation.
Manufacturing	Production orders with BOM explosion, routing operations, component issues, finished goods receipts, WIP costing (material/labor/overhead), and variance settlement.
Bank Reconciliation	Statement line matching against cleared payments, outstanding checks, deposits in transit, bank fees, and zero-net-difference enforcement.
Financial Reporting	Balance sheet, income statement, cash flow statement (indirect method), and changes in equity — derived from trial balance with prior-period comparatives, management KPIs, and budget variance analysis.
Banking Operations	Customer onboarding, KYC reviews, account management, transaction lifecycle (authorize/complete/reverse), suspicious activity flagging, and account freezing — with full OCEL 2.0 event coverage.
Audit Lifecycle	Engagement planning, risk assessment, workpaper creation and review, evidence collection, finding management, remediation tracking, and professional judgment documentation — ISA-aligned OCPM events.
Tax Accounting	Tax jurisdictions (Federal/State/Local/Supranational), tax code master data, tax line decoration on source documents (VI/CI/JE/Payment/Payroll), VAT/GST/income/withholding tax returns, ASC 740/IAS 12 provision computation with deferred tax, FIN 48/IFRIC 23 uncertain positions, cross-border withholding with treaty benefits.
Treasury & Cash Mgmt	Daily cash positions, probability-weighted forecasts, cash pooling (physical/notional/zero-balance sweeps), FX forwards, IR swaps, and options with ASC 815/IFRS 9 hedge effectiveness testing, debt instruments with covenants & amortization, bank guarantees, multilateral netting.
Project Accounting	WBS hierarchies, cost lines (labor/material/subcontractor/overhead/equipment/travel), percentage-of-completion revenue recognition (ASC 606), earned value management (BCWS/BCWP/ACWP/SPI/CPI/EAC/ETC/TCPI), milestones with payment tracking, change orders, retainage holds and releases.
ESG / Sustainability	GHG Protocol Scope 1/2/3 emissions (activity-based & spend-based), energy consumption with renewable targets, water/waste management, workforce diversity & pay equity, safety incidents (TRIR/LTIR/DART), governance metrics, GRI/SASB/TCFD disclosure frameworks, supplier ESG assessments, climate scenario analysis. 8

Each chain defaults to **enabled: false** and integrates with the existing generation pipeline via new orchestrator phases. All new fields on existing structs use **Option<>** for backward compatibility.

2.4 Country Pack Architecture

New in v0.8.0, expanded in v0.11.0 — pluggable JSON-based country configuration replacing ~7500 lines of hardcoded country-specific data:

- **10 built-in packs:** US, DE, GB, FR, JP, CN, IN, IT, ES, CA with 16-section JSON schema covering locale, names, holidays, tax rates, addresses, phone formats, banking, payroll rules, legal entity formats, and document texts.
- **Layered merge:** `_default.json` → country pack → user overrides — objects merge recursively, arrays/scalars replace, enabling surgical per-country customization.
- **5 holiday types, 15 regions:** Fixed dates, Easter-relative, nth-weekday (with offset), last-weekday, and lunar calendar algorithms (Chinese New Year, Diwali, Eid al-Fitr, Chuseok). Holiday calendars for US, DE, GB, FR, IT, ES, CA, CN, JP, IN, BR, MX, AU, SG, KR.
- **Generator integration:** `HolidayCalendar`, `MultiCultureNameGenerator`, tax code, payroll, emission, and customer generators all consume country pack data — wired through `EnhancedOrchestrator` per company.
- **Extensibility:** `country_packs.external_dir` config loads custom or commercial packs from disk; no code changes required to add a new country.

2.5 Banking, KYC & AML

A dedicated banking module generates realistic transaction data for anti-money-laundering testing:

- **Customer personas:** Retail, Business, Trust profiles with full KYC envelopes (declared turnover, source of funds, geographic exposure, cash intensity).
- **AML typologies:** Structuring, funnel accounts, layering schemes, money mule networks, round-tripping, and adversarial spoofing for robustness testing.
- **Ground-truth labels:** Entity-level risk classifications, transaction-level labels, and investigation narratives.

2.6 Audit Simulation

Generates ISA-compliant audit artifacts:

- Engagement metadata with materiality thresholds (ISA 320).
- Workpapers per ISA 230, evidence per ISA 500.
- Risk assessments (ISA 315/330), findings (ISA 265), and professional judgment documentation (ISA 200).
- *New in v0.11.0:* External confirmations per ISA 505 — AR (40%), AP (30%), Bank (20%), Legal (10%) with positive/negative response modeling, reconciliation for disagreements, and alternative procedures for non-responses.

2.7 YAML-Driven Audit FSM Engine

New in v2.0.0 — DataSynth introduces a methodology-driven approach to audit data generation through the `datasynth-audit-fsm` crate. Rather than generating audit artifacts through independent, stateless generators, the engine loads audit methodology blueprints—encoded as YAML-defined finite state machines—and walks each procedure’s aggregate state machine in topological order, respecting precondition dependencies.

2.7.1 Two-Layer Architecture

The engine separates *what happens* (methodology blueprint) from *how it generates* (generation overlay):

- **Blueprints** define procedures, phases, state transitions, evidence requirements, actors, and standards references. Twelve built-in blueprints ship with v2.0.0, spanning ISA, IIA-GIAS, PCAOB, AICPA SOC 2, and Big 4 firm-style methodologies.
- **Overlays** control revision probability, timing distributions, artifact volumes, anomaly injection rates, and actor behavior profiles. Three presets are provided: `default`, `thorough`, and `rushed`.
- **Judgment levels** — each step carries a `judgment_level` classification (`data_only`, `ai_assistable`, or `human_required`), supporting analysis of automation potential across methodologies.

Blueprint	Framework	Procedures	Steps	Standards
Generic FSA	ISA	9	24	14
Generic IA	IIA-GIAS	34	82	52
Deloitte / PwC / KPMG FSA	ISA	10	28	16
Deloitte / PwC / KPMG ISA (full)	ISA	48	180	34
EY GAM Lite	ISA	12	36	18
PCAOB Integrated	PCAOB	42	156	28
SOC 2 Type II	AICPA TSC	20	60	15
Regulatory Exam	Regulatory	16	48	12

2.7.2 Key Capabilities

- **Event-sourced audit trail** — every state transition and procedure step emits a deterministic audit event with actor, timestamp, command, evidence references, and standards citations.
- **Concrete artifact generation** — a step dispatcher maps 135 blueprint commands to 14 pre-initialized generators, producing typed artifacts: materiality calculations, risk assessments, workpapers, evidence, sampling plans, analytical procedures, audit opinions, and more.
- **8-state C2CE lifecycle** — the Condition–Criteria–Cause–Effect finding development workflow models realistic audit finding progression.
- **Self-loop handling** — follow-up procedures with re-verification cycles are bounded by configurable iteration limits.
- **Continuous phases** — governance, ethics, and quality phases execute in parallel with sequential audit phases.
- **Discriminator filtering** — scope engagements by category (financial, operational, compliance, IT), risk rating, or engagement type.
- **OCEL 2.0 projection** — audit events map to the Object-Centric Event Log standard for integration with process mining tools.

- **Streaming execution** — callback-based API emits each audit event in real time during FSM traversal, enabling downstream consumption without buffering the full engagement result.
- **ISA 600 group audit** — component auditor instructions, reports, and multi-entity coordination for group audit simulation.
- **Year-over-year engagement chains** — multi-period audit simulation with prior-year comparative procedures and opening balance reliance.

2.7.3 Audit Path Optimizer

The companion `datasynth-audit-optimizer` crate converts methodology blueprints into directed graphs for analytical queries:

- **Shortest path analysis** — minimum transitions to completion (FSA: 27, IA: 101).
- **Constrained path** — must-visit procedures with transitive precondition expansion.
- **Monte Carlo simulation** — stochastic outcome distributions across N engagement simulations, identifying bottleneck procedures, revision hotspots, and the statistically most common completion path.

2.8 COSO 2013 Internal Control Framework

Full integration with the COSO Internal Control-Integrated Framework:

- **5 Components:** Control Environment, Risk Assessment, Control Activities, Information & Communication, Monitoring Activities.
- **17 Principles:** Complete principle coverage with control mappings.
- **Control Scopes:** Entity-level, transaction-level, IT general, and IT application controls.
- **Maturity Levels:** 6-level model (Non-Existent through Optimized).
- **Export:** `coso_control_mapping.csv` with principle-level granularity.

2.9 Accounting & Audit Standards

Comprehensive standards support via the `datasynth-standards` crate:

Category	Standards Supported
Accounting (US GAAP)	ASC 606 (Revenue), ASC 842 (Leases — 5 bright-line classification tests), ASC 820 (Fair Value), ASC 360 (Impairment)
Accounting (IFRS)	IFRS 15 (Revenue), IFRS 16 (Leases — enhanced ROU measurement), IFRS 13 (Fair Value), IAS 36 (Impairment)
Accounting (French GAAP)	Plan Comptable Général 2024 (PCG) chart of accounts, FEC export (Article A47 A-1, 18 columns), auxiliary GL sub-accounts (401/411 series)
Accounting (German GAAP)	HGB §§238–263, SKR04 chart of accounts (classes 0–9), GoBD audit export (13-column journal + XML index), Degressiv depreciation (EStG §7(2)), GWG low-value asset expensing (§6(2), ≤800 EUR)
Audit Standards	34 ISA standards (ISA 200–720), 19+ PCAOB standards with ISA mapping
Regulatory	SOX Section 302 (Certifications), SOX Section 404 (ICFR Assessment), Deficiency Matrix (MW/SD classification)

Supports dual reporting mode (US GAAP + IFRS) with automatic framework difference reconciliation. A generalized **FrameworkAccounts** struct maps ~45 semantic account purposes to framework-specific GL codes, enabling consistent multi-GAAP data generation with a single code path per framework.

2.10 Process Mining (OCEL 2.0 & XES 2.0)

Major expansion in v0.7.0 — Object-Centric Event Logs now cover all 12 enterprise process families with 101+ activity types and 65+ object types, enabling cross-process conformance checking and end-to-end variant analysis.

- **OCEL 2.0:** JSON and XML export with objects, events, and object relationships.
- **XES 2.0:** IEEE standard format for ProM, Celonis, Disco, and pm4py.
- **12 Process Families:** P2P (9), O2C (10), R2R/A2R (17), S2C (8), H2R (8), MFG (10), BANK (8), AUDIT (10), Tax (8), Treasury (4), ProjectAccounting (5), ESG (3) — 101+ activities total.
- **65+ Object Types:** Lifecycle states and relationships across all families — from purchase orders through ESG disclosures.
- **3 Variant Types:** HappyPath (75%), ExceptionPath (20%), ErrorPath (5%) per generator for realistic process deviation.
- **Deterministic UUIDs:** FNV-1a hash-based IDs with type discriminators for reproducible event logs.

3 AI-Augmented Generation

New in v0.5.0 — three opt-in AI backends that enhance the rule-based core without sacrificing determinism or reproducibility.

3.1 LLM-Augmented Metadata Enrichment

An `LlmProvider` trait integrates external language models for contextual metadata enrichment while preserving deterministic fallback:

- **Provider trait:** `MockProvider` (deterministic, default) and `HttpProvider` (OpenAI, Anthropic, or custom API) implementations.
- **Enrichment targets:** Vendor names, transaction descriptions, anomaly explanations, and memo fields.
- **Natural language config:** Generate YAML configurations from plain English descriptions.
- **Response caching:** LRU cache with deterministic fallback ensures reproducibility even with stochastic providers.

3.2 Diffusion Model Integration

A statistical diffusion backend learns distribution shapes from fingerprints and blends them with rule-based outputs:

- **Noise schedules:** Cosine, linear, and sigmoid with configurable step count.
- **Hybrid mode:** Weighted blend of rule-based and diffusion-generated values (`diffusion_weight` 0.0–1.0).
- **Training pipeline:** `DiffusionTrainer` with epoch count, batch size, and learning rate configuration.

3.3 Causal & Counterfactual Generation

Structural causal models enable “what-if” scenario generation grounded in do-calculus:

- **CausalGraph:** DAG specification of causal relationships between entities (e.g., vendor risk → invoice amount → approval level).
- **StructuralCausalModel:** Functional causal model with noise terms and topological evaluation.
- **Interventions:** Do-calculus interventions for hypothetical scenarios (“what if vendor risk doubled?”).
- **Counterfactuals:** Generate alternative versions of existing records under modified assumptions.
- **Templates:** Pre-built causal graphs for fraud detection, revenue impact analysis, and supply chain modeling.

Major expansion in v0.10.0 — a full **Counterfactual Simulation Engine** for paired baseline/counterfactual dataset generation:

- **CausalDAG:** 16-node default financial process DAG with 8 transfer function types (Linear, Exponential, Logistic, Step, Threshold, Decay, Piecewise, InverseLogistic) and forward propagation with lag support.

- **8 InterventionTypes:** ParameterShift, MacroShock, ControlFailure, EntityEvent, ProcessChange, RegulatoryChange, Composite, and Custom — each with timing, onset interpolation, and priority-based conflict resolution.
- **ConfigMutator:** Dot-path config mutation for applying causal propagation results to generation parameters.
- **DiffEngine:** Baseline vs. counterfactual comparison with summary, record-level, and aggregate diff formats including KPI impact analysis.
- **CLI:** `scenario {list, validate, generate, diff}` subcommands for end-to-end scenario management.

3.4 Unified Generation Pipeline

New in v0.11.0 — stateful multi-period generation with fraud scenario packs, phase-aware streaming, and enriched OCEL 2.0 event logs:

- **GenerationSession:** Stateful multi-period generation that splits output into fiscal-year-aligned periods, carrying forward balances, entity state, and document IDs between periods via `.dss` checkpoint files. Supports incremental generation (`-append -months N`).
- **5 Fraud Scenario Packs:** Pre-built YAML configurations (`revenue_fraud`, `payroll_ghost`, `vendor_kickback`, `management_override`, `comprehensive`) applied via deep-merge with `-fraud-scenario` and `-fraud-rate` CLI flags.
- **StreamPipeline:** Phase-aware streaming with file, HTTP, and no-op targets. JSONL envelope format with generation phase metadata for feeding downstream pipelines during generation.
- **OCEL 2.0 Enrichment:** Lifecycle state machines (PO, SO, VI with typed state transitions), multi-object correlation events (three-way match, payment allocation, bank reconciliation), and resource pool workload modeling with round-robin, least-busy, and skill-based assignment strategies.

4 Machine Learning & Analytics

4.1 ACFE-Aligned Fraud Taxonomy

DataSynth implements the ACFE (Association of Certified Fraud Examiners) “Report to the Nations” taxonomy with calibrated statistics:

ACFE Category	Prevalence	Scheme Types
Asset Misappropriation	86%	20+ cash schemes: skimming, larceny, shell company, ghost employee, billing fraud, expense reimbursement
Corruption	33%	Conflicts of interest, bribery, kickbacks, bid rigging, economic extortion
Financial Statement	10%	Revenue manipulation, expense timing, concealed liabilities, improper disclosures

Calibration: \$117k median loss, 12-month median duration, detection method distribution (42% tips, 16% internal audit, 12% management review).

4.2 Advanced Fraud Modeling

- **Collusion & Conspiracy:** 9 ring types (EmployeePair, DepartmentRing, EmployeeVendor, VendorRing), 6 conspirator roles, defection modeling based on detection risk.
- **Management Override:** Executive-level fraud with override techniques, Fraud Triangle modeling (Pressure, Opportunity, Rationalization), concealment strategies.
- **Red Flag Patterns:** 40+ patterns with Bayesian probabilities, 6 categories (Vendor, Transaction, Timing, Approval, Document, Behavioral), realistic false positive rates.
- **Industry-Specific:** Manufacturing (yield manipulation, phantom production), Retail (sweethearting, refund fraud), Healthcare (upcoding, unbundling, phantom billing).
- **Procurement Fraud:** Bid rigging, phantom vendor contracts, split-to-avoid-threshold, conflict of interest, maverick spend, expired contract purchases, qualification bypass.
- **HR/Payroll Fraud:** Ghost employee payroll, payroll inflation, duplicate expense reports, fictitious expenses, split expense to avoid approval.

Anomaly injection is **entity-aware**: injection rates are adjusted per vendor, employee, and account based on contextual risk factors (e.g., new vendors, high-value accounts, employees with recent access changes). Every injected anomaly carries a **LabeledAnomaly** record with full meta-data, enabling supervised and semi-supervised learning pipelines without manual labeling effort.

4.3 Data Quality Variations

Realistic data imperfections for training data-quality ML models:

- **Missing values:** MCAR, MAR, MNAR, and systematic patterns.
- **Format variations:** Date, amount, and identifier format diversity across regional conventions.
- **Duplicates:** Exact, near, and fuzzy duplicates.
- **Typos:** Keyboard-aware substitution, OCR errors, homophones.
- **Encoding issues:** Mojibake, BOM artifacts, HTML entity corruption.

4.4 Enterprise Interconnectivity

Realistic enterprise network modeling with configurable relationships:

- **Multi-Tier Vendor Networks:** 3-tier supply chain (Tier1/Tier2/Tier3), vendor clusters (ReliableStrategic 20%, StandardOperational 50%, Transactional 25%, Problematic 5%), lifecycle stages, quality scores, dependency analysis.
- **Customer Value Segmentation:** Enterprise (40% revenue/5% customers), MidMarket (35%/20%), SMB (20%/50%), Consumer (5%/25%), lifecycle stages, referral networks, corporate hierarchies.
- **Entity Relationship Graph:** 51+ entity types across 10 process families, 28+ relationship types with typed edge constraints, composite strength calculation from volume, count, duration, recency, and mutual connections.
- **Cross-Process Links:** P2P↔O2C linkage via inventory (GoodsReceipt→Delivery), inter-company bilateral links.

4.5 Pattern & Process Drift

Expanded in v0.11.0 — comprehensive drift modeling with dedicated generators for realistic temporal evolution:

- **Organizational Events:** `OrganizationalEventGenerator` produces Acquisition (1.35x volume, 5% integration errors), Divestiture, Reorganization, LeadershipChange, WorkforceReduction, and Merger events with multi-phase integration timelines and volume/error-rate multipliers.
- **Process Evolution:** `ProcessEvolutionGenerator` with S-curve automation rollout, approval workflow transitions between `WorkflowType` variants, policy changes, and control enhancements.
- **Drift Event Meta-Generator:** `DriftEventGenerator` observes organizational and process events to produce ML ground-truth `LabeledDriftEvent` records with drift type (Statistical, Temporal, Behavioral, Market, Organizational, Process), magnitude, affected fields, and detection difficulty.
- **Technology Transitions:** ERP migration phases with error rate and processing time multipliers, duplicate/missing/format issues.
- **Behavioral Drift:** Vendor payment term extensions, customer delays during downturns, employee learning curves, year-end intensity.
- **Market Drift:** Economic cycles (sinusoidal, asymmetric, mean-reverting), industry-specific cycles, commodity drift, price shocks.

4.6 Graph & Network Export

Supported Graph Formats

PyTorch Geometric	.pt files with node features, edge index, edge attributes, labels, and train/val/test masks.
Neo4j	CSV node/edge files with Cypher import scripts.
DGL	Deep Graph Library format for GNN training.
RustGraph	Native Rust graph format with adjacency lists.
RustGraph Hyper	Multi-layer hypergraph export (governance, process event, and accounting network layers with cross-layer edges).

Expanded in v0.7.0 — the multi-layer hypergraph now spans all 8 process families with 24 entity type codes (100–400), per-family document nodes (S2C, H2R, MFG, BANK, AUDIT, Bank Recon), and OCPM events as hyperedges connecting all participating objects when `events_as_hyperedges` is enabled.

All five exporters share a unified `CommonExportConfig` for consistent feature engineering, label selection, and train/val/test mask settings. Computed features include temporal signals (week-day, period-end flags), amount signals (log-amount, Benford probability, round-number flag), structural signals (line count, unique accounts), and one-hot categorical encodings.

5 Privacy & Fingerprinting

5.1 Privacy-Preserving Fingerprint Extraction

DataSynth can extract a statistical *fingerprint* from real data and synthesize new data that matches its properties — without ever copying individual records.



5.2 Privacy Levels

Level	Epsilon (ϵ)	k-Anon	Outlier %	Use Case
Minimal	5.0	3	99%	Low privacy, high utility
Standard	1.0	5	95%	Balanced (default)
High	0.5	10	90%	Sensitive environments
Maximum	0.1	20	85%	Maximum privacy guarantees

The privacy engine combines differential privacy (Laplace and Gaussian mechanisms), k-anonymity with rare-value suppression, and outlier winsorization. A full privacy audit trail records every decision and the cumulative epsilon budget spent.

5.3 Federated Fingerprinting & Certificates

New in v0.5.0 — distributed fingerprint extraction and cryptographic provenance:

- **Federated Coordinator:** Aggregate fingerprints from distributed data sources without centralizing raw data.
- **Synthetic Data Certificates:** `SyntheticDataCertificate` with cryptographic proof of differential privacy guarantees, generation metadata, and validation workflow.
- **Pareto Frontier:** Automated exploration of optimal privacy–utility trade-offs across epsilon values.

5.4 Formal DP Composition

New in v0.5.0 — advanced privacy accounting beyond naive ϵ -summation:

- **Rényi DP:** RDP curve tracking at α values 2–128 with conversion to (ϵ, δ) -DP.
- **zCDP:** Zero-Concentrated DP with additive ρ composition and tighter bounds.
- **Budget Manager:** Global budget tracking across extraction runs with composition-aware enforcement.
- **Privacy Evaluation:** Post-generation membership inference attack (MIA) testing, linkage attack assessment, and NIST SP 800-226 alignment.

5.5 Fidelity Evaluation

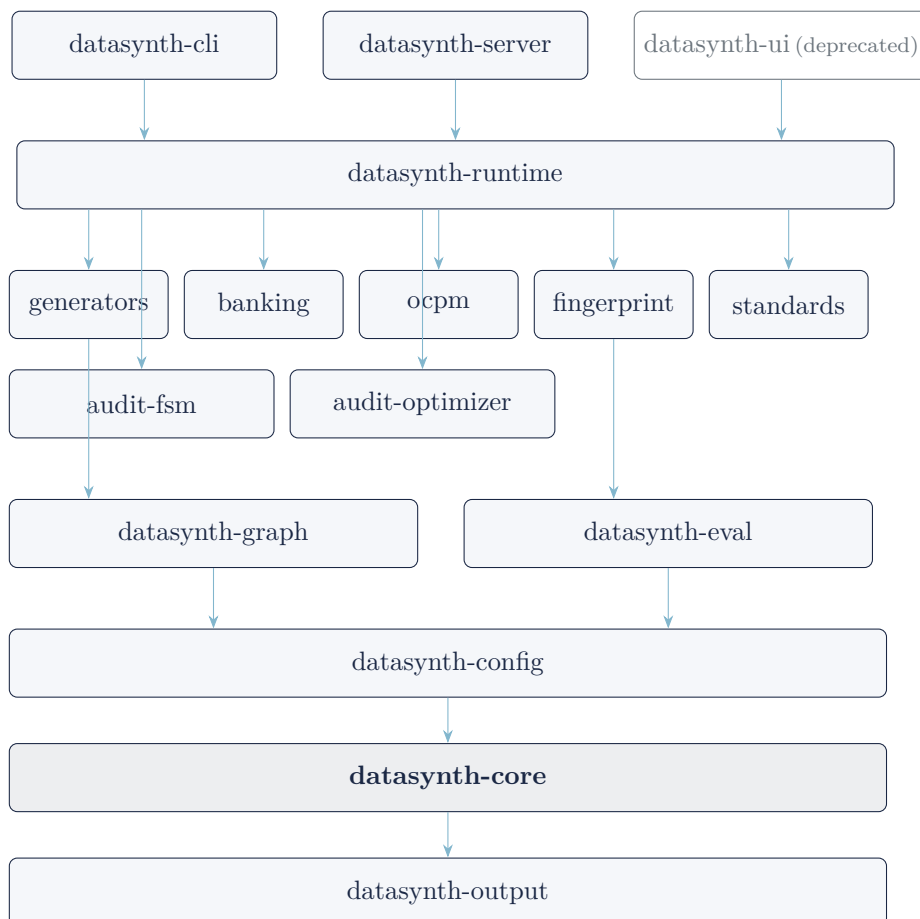
Per-column statistical distance metrics quantify how well synthetic data matches the source fingerprint:

Metric	Description
KS Statistic	Kolmogorov-Smirnov two-sample test statistic (sup of CDF difference).
Wasserstein-1	Earth Mover's Distance via piecewise-linear inverse CDF integration over 9 percentile knots.
JS Divergence	Jensen-Shannon divergence from percentile-bin PMFs, bounded by $\ln 2$.

Distribution CDFs are computed analytically for Normal, LogNormal, Gamma (regularized incomplete gamma via Lanczos + Lentz continued fraction), Pareto, PointMass, and Mixture distributions.

6 Architecture

6.1 Layered Crate Architecture



6.2 Production-Grade Infrastructure

Capability	Details
Deterministic Output	ChaCha8 PRNG with configurable seed; identical seed $\Rightarrow$ identical dataset.
Financial Precision	<code>rust_decimal</code> throughout; no IEEE 754 floating-point artifacts. Decimals serialized as strings.
Resource Guards	Unified CPU, memory, and disk monitoring with automatic throttling and graceful degradation (Normal $\rightarrow$ Reduced $\rightarrow$ Minimal $\rightarrow$ Emergency).
Collision-Free IDs	FNV-1a hash-based UUID factory with generator-type discriminators prevents document ID collisions across parallel generators.
Streaming Output	CSV, JSON, NDJSON, Apache Parquet (15-column Arrow schema, Zstd compression), and XBRL 2.1 instance documents (US GAAP & IFRS taxonomies). Streaming sinks with backpressure strategies (Block, DropOldest, DropNewest, Buffer).
API Layer	REST + gRPC + WebSocket with Argon2id API-key auth, JWT/OIDC bearer tokens (RS256, Keycloak/Auth0/Entra ID), RBAC (Admin/Operator/Viewer), sliding-window rate limiting (in-memory or Redis), TLS (rustls), async job queue, security headers, and structured audit logging.
Desktop UI	<i>Deprecated in v2.2.0.</i> Previously a Tauri + SvelteKit application; superseded by REST/gRPC API and Python wrapper workflows.
Plugin SDK	<code>GeneratorPlugin</code> , <code>SinkPlugin</code> , and <code>TransformPlugin</code> traits with thread-safe registry for custom extensions.
Webhooks	Fire-and-forget event dispatch (RunStarted, RunCompleted, RunFailed, GateViolation) with HMAC signing and configurable retry.
Data Lineage	Per-file SHA-256 checksums, <code>LineageGraph</code> tracking config $\rightarrow$ generator $\rightarrow$ output relationships, W3C PROV-JSON export, CLI <code>verify</code> command.
Python Wrapper	<code>datasynth-py</code> 1.10 with blueprints (sourcing, HR, manufacturing, financial reporting, tax, treasury, project accounting, ESG), pandas/polars integration, async client, and WebSocket streaming support.

7 Industry Presets

DataSynth ships with ten industry-specific configuration presets, each tuned for realistic business process weightings, chart-of-accounts structures, and regional multi-company setups.

Industry	Key Weight	Characteristics
Manufacturing	40% P2P	14 transaction types, multi-level BOM, production routings, work centers, JIT, quality framework.
Retail	50% O2C	12 transaction types, 7 store types, shrinkage modeling, loss prevention, markdown patterns.
Healthcare	15% H2R	15 transaction types, payer mix (Medicare/Medicaid/Commercial), ICD-10/CPT/DRG coding, HIPAA/Stark compliance.
Financial Services	40% R2R	Report-heavy, regulatory compliance, intercompany structures, loan loss provisions.
Technology	15% H2R	SaaS revenue recognition, R&D capitalization, deferred revenue.
Professional Svcs.	—	Time-based billing, project accounting, trust accounting.
Energy	—	Capital-intensive, long-lived assets.
Transportation	—	Fleet management, route-based costing.
Real Estate	—	Property portfolios, lease accounting.
Telecommunications	—	Subscription revenue, network assets.

Each preset supports three complexity tiers: **Small** (~100 GL accounts), **Medium** (~400 accounts), and **Large** (~2 500 accounts).

8 Statistical Distribution Framework

DataSynth includes an advanced statistical distribution framework for generating empirically grounded synthetic data:

Capability	Details
Mixture Models	Gaussian and Log-Normal mixtures with weighted components (“routine”, “significant”, “major” transactions).
Copula Correlation	Gaussian, Clayton, Gumbel, Frank, Student-t copulas for cross-field dependencies.
Distribution Types	Pareto (heavy-tailed), Weibull (time-to-event), Beta (proportions), Zero-inflated (excess zeros).
Benford’s Law	First and second-digit compliance, round number bias, threshold clustering.
Regime Changes	Economic cycles, acquisition/divestiture effects, recession modeling.
Industry Profiles	Pre-configured profiles for Retail, Manufacturing, Financial Services.

All distributions include validation tests: Benford’s Law MAD, Anderson-Darling goodness-of-fit, chi-squared, and correlation matrix verification.

9 Use Cases

Primary Use Cases

- ▶ **Fraud Detection ML** — Train and validate models on labeled fraud typologies with realistic base-rate imbalance.
- ▶ **Graph Neural Networks** — Export transaction graphs in PyTorch Geometric, Neo4j, DGL, or RustGraph format — including multi-layer hypergraphs — with pre-computed features and train/val/test splits.
- ▶ **AML & KYC Testing** — Generate banking transaction data with structuring, layering, and mule patterns for compliance system validation.
- ▶ **Audit Analytics** — Produce ISA-compliant audit artifacts and anomaly-injected financial data for analytics tool development.
- ▶ **ERP Integration Testing** — Generate SAP ACDOCA-format data with full document chains for system migration and integration testing.
- ▶ **Process Mining** — OCEL 2.0 event logs for process discovery, conformance checking, and variant analysis.
- ▶ **SOX & COSO Compliance Testing** — Internal control definitions with COSO 2013 mappings, SOX 302/404 certifications, deficiency classification, and SoD conflict detection.
- ▶ **Accounting Standards Testing** — Generate revenue contracts (ASC 606/IFRS 15), lease portfolios (ASC 842/IFRS 16), fair value measurements, and impairment tests with dual-framework reconciliation.
- ▶ **Data Quality ML** — Labeled missing values, typos, duplicates, and format variations for training data-cleansing models.
- ▶ **Drift Detection ML** — Labeled drift events (organizational, process, technology, market, regulatory) with magnitude and detection difficulty for drift detection model training.
- ▶ **Network Analytics** — Multi-tier vendor networks and customer segmentation graphs with relationship strength metrics for supply chain and customer analytics.
- ▶ **Causal Inference Testing** — Structural causal models with do-calculus interventions for counterfactual scenario generation and what-if analysis.
- ▶ **Pipeline Orchestration** — Apache Airflow operators, dbt source generation, MLflow experiment tracking, and Spark DataFrame readers for end-to-end ML pipelines.
- ▶ **Tax Compliance Testing** — Tax return filing simulation, ASC 740/IAS 12 provision computation, uncertain tax position modeling, withholding tax treaty benefits.
- ▶ **Treasury Operations** — Cash positioning, probability-weighted forecasting, hedge effectiveness testing (ASC 815/IFRS 9), debt covenant monitoring, and multilateral netting.
- ▶ **Project Cost Control** — WBS-based project costing, earned value management (EVM), change order tracking, percentage-of-completion revenue recognition, retainage.
- ▶ **ESG & Sustainability Reporting** — GHG Protocol Scope 1/2/3 emissions, diversity & pay equity metrics, GRI/SASB/TCFD disclosure preparation, supplier ESG assessments.
- ▶ **EU AI Act Compliance** — Article 50 synthetic content marking with content credentials and Article 10 data governance reports with bias assessments.

10 Evaluation & Auto-Tuning

DataSynth includes a built-in evaluation framework that measures synthetic data quality across four dimensions:

1. **Statistical Fidelity** — KS statistic, Wasserstein-1 (Earth Mover’s Distance), Jensen-Shannon divergence, Benford’s Law MAD, Anderson-Darling goodness-of-fit.
2. **Coherence** — Balance-sheet validation, intercompany matching, document chain integrity, standards compliance.
3. **Data Quality** — Completeness, consistency, duplicate rates, format correctness, uniqueness.
4. **ML Readiness** — Feature distributions, label quality, graph structure, train/val/test split balance.
5. **Drift Detection** — Hellinger distance, Population Stability Index (PSI), rolling window analysis, detection delay metrics.

10.1 ML Benchmarks

Pre-configured benchmarks aligned with ACFE statistics and industry verticals:

- **ACFE-Calibrated:** General fraud (1k), collusion-focused (5k), management override (2k).
- **Industry-Specific:** Manufacturing fraud (5k), retail fraud (10k), healthcare fraud (5k), technology fraud (3k), financial services fraud (5k).
- **Cost-Sensitive:** Asymmetric cost matrices for realistic fraud detection evaluation.

An **auto-tuning engine** analyses evaluation results and produces prioritized configuration patches with expected improvement estimates — closing the loop between generation and validation.

10.2 Quality Gate Engine

New in v0.5.0 — configurable pass/fail thresholds that can block releases of low-quality synthetic data:

- **8 metrics:** Benford MAD, balance coherence, document chain completeness, duplicate rate, missing rate, distribution fit, correlation accuracy, anomaly precision.
- **Built-in profiles:** `strict`, `default`, `lenient` — or define custom per-metric thresholds.
- **CLI integration:** `-quality-gate <profile>` with exit code 2 on failure for CI/CD pipelines.

11 Compliance & Governance

11.1 EU AI Act

DataSynth implements Article 50 (synthetic content marking) and Article 10 (data governance) requirements:

- **Content Credentials:** `SyntheticContentMarker` creates machine-readable credentials (generator name, version, timestamp, config hash) in embedded, sidecar, or both formats.
- **Data Governance Report:** `DataGovernanceReport` with `BiasAssessment` documenting data provenance, quality metrics, and bias evaluation for regulatory submission.

11.2 Regulatory Compliance Documentation

Pre-built compliance guides with self-assessment templates:

Framework	Coverage
EU AI Act	Article 50 content marking, Article 10 governance
NIST AI RMF	Self-assessment across MAP, MEASURE, MANAGE, GOVERN
GDPR	Article 30 records, DPIA guidance, data minimization
SOC 2 Type II	Readiness across 5 Trust Service Criteria
ISO 27001:2022	Annex A alignment (11 implemented, 6 partial)

12 DevOps & Deployment

12.1 CI/CD Pipeline

An 8-job GitHub Actions pipeline ensures release quality:

1. **fmt**: `cargo fmt -check`
2. **clippy**: Lint with `-D warnings`
3. **test**: Cross-platform matrix (Ubuntu, macOS, Windows)
4. **eval**: Comprehensive evaluation (23 evaluators, all process families — main branch only)
5. **msrv**: Minimum Supported Rust Version validation (1.88)
6. **security**: `cargo deny + cargo audit`
7. **coverage**: `cargo-llvm-cov` with Codecov
8. **benchmarks**: Criterion regression checks on PRs

12.2 Deployment Options

Target	Details
Docker	Multi-stage builds with cargo-chef caching; distroless runtime images for server and CLI.
Docker Compose	Server + Prometheus + Grafana stack with auto-provisioned data-sources.
Kubernetes	Helm chart with HPA (2–10 replicas), PDB, rolling updates, optional Redis subchart, Prometheus ServiceMonitor, and ConfigMap/Secret templates.
Bare Metal	SystemD service with security hardening (NoNewPrivileges, ProtectSystem, MemoryMax=4G, CPUQuota=200%).
Release Pipeline	Pre-built binaries for 5 platforms, Docker image to GHCR (amd64 + arm64), Trivy container scanning.

12.3 Ecosystem Integrations

Python-native integrations for popular data engineering tools:

- **Apache Airflow:** `DataSynthOperator`, `DataSynthSensor`, and `DataSynthValidateOperator` for orchestrated generation.
- **dbt:** `DbtSourceGenerator` for source YAML and seed files.
- **MLflow:** `DataSynthMlflowTracker` for experiment tracking with quality metrics.
- **Apache Spark:** `DataSynthSparkReader` for direct `DataFrame` loading from DataSynth output.

13 Getting Started

13.1 Quick Start (CLI)

```
# Generate with demo preset
datasynth-data generate -demo -output ./output

# Create an industry-specific config
datasynth-data init -industry manufacturing -complexity medium -o config.yaml

# Generate from config
datasynth-data generate -config config.yaml -output ./output
```

13.2 Quick Start (Python)

```
from datasynth_py import DataSynth
from datasynth_py.config import blueprints

config = blueprints.retail_small(companies=4, transactions=10000)
synth = DataSynth()
result = synth.generate(config=config)
```

13.3 Server Mode

```
# Start REST/gRPC server with 4 worker threads
cargo run -p datasynth-server - -port 3000 -worker-threads 4
```

Links & Resources

Repository	https://github.com/mivertowski/SyntheticData
Documentation	https://mivertowski.github.io/SyntheticData/
Crates.io	https://crates.io/crates/datasynth-core
PyPI	https://pypi.org/project/datasynth-py/